

Task 3: Model Validation, Overfitting Control & Hyperparameter Tuning

Maincrafts Technology — AI & ML Internship | California Housing Dataset

1. Overfitting Analysis

Overfitting happens when a model learns the training data too well — including its noise — and then performs poorly on data it hasn't seen before. The most obvious sign is a large gap between training performance and test performance.

In this task, an unconstrained Decision Tree was first trained with no restrictions on depth. The result was a near-perfect training RMSE (essentially memorizing the data) but a significantly higher test RMSE. This is a clear case of overfitting — the model had too much capacity relative to the signal in the data.

What Was Observed:

Metric	Train RMSE	Test RMSE	Gap
Decision Tree (no constraint)	~0.00	~0.72	Large — overfitting
Decision Tree (tuned)	~0.55	~0.63	Small — controlled

To visualize this, the model was trained at increasing max_depth values (1 through 19). As depth increased, training RMSE kept dropping while test RMSE eventually started rising — the textbook bias-variance tradeoff curve. The sweet spot is where test RMSE is lowest, which the tuned model targets.

2. Cross-Validation Approach

A single train-test split is convenient but unreliable. Depending on which samples end up in test vs train, your measured accuracy can shift noticeably. If you tune your model based on a lucky split, you're not actually improving the model — you're just fitting to that particular split.

5-Fold Cross-Validation addresses this by splitting the full dataset into 5 equal parts. The model is trained 5 times, each time using a different fold as the validation set and the remaining 4 as training. The final score is the average across all 5 — a much more honest estimate of real performance.

CV Results Summary:

Model	5-Fold CV RMSE
Linear Regression	~0.73
Ridge Regression	~0.73
Decision Tree (unconstrained)	~0.72

Cross-validation is used before deployment in professional settings because it reflects how a model will behave on data it hasn't seen — which is the only thing that actually matters in production.

3. Hyperparameter Tuning

Hyperparameters are settings you choose before training — they control the model's behavior rather than being learned from data. Picking good values is not guesswork; GridSearchCV automates the process by trying every combination in a defined grid and using cross-validation to score each one.

Decision Tree — Parameter Grid:

`max_depth`: [3, 5, 7, 10] — controls how deep the tree can grow. Shallower trees generalize better but may underfit. `min_samples_split`: [2, 5, 10] — minimum samples needed to make a split. Higher values prevent the tree from making splits on very small groups, which reduces overfitting.

Ridge Regression — Parameter Grid:

`alpha`: [0.01, 0.1, 1.0, 10.0, 100.0] — controls regularization strength. Higher alpha shrinks coefficients more aggressively, reducing overfitting but potentially increasing bias.

GridSearchCV uses 5-fold CV internally for each combination, so the best parameters are chosen based on average validation performance — not just performance on one fold.

4. Final Model Comparison

Model	Train RMSE	Test RMSE	R ² Score	Overfit Gap
Linear Regression	~0.72	~0.73	~0.60	Low
Ridge (Tuned)	~0.72	~0.73	~0.60	Low
Decision Tree (Tuned)	~0.55	~0.63	~0.72	Moderate

5. Final Model Selection Justification

The tuned Decision Tree was selected as the final model based on test RMSE and R² score. Here is the reasoning:

Why this model: The tuned Decision Tree achieved the best R² (~0.72) among all three, meaning it explains more variance in house prices than the linear models. The test RMSE is also the lowest after tuning.

How overfitting was reduced: Constraining `max_depth` to the GridSearch-optimal value and increasing `min_samples_split` prevents the tree from making overly specific splits. The train-test RMSE gap dropped from massive (unconstrained) to manageable (tuned).

Why CV results are trusted: The best parameters were chosen based on 5-fold cross-validation performance inside GridSearchCV, not just test set performance. This means the model selection process

didn't peek at the test set — so test results are a genuinely unseen evaluation.

Trade-offs considered: Linear Regression is simpler, more interpretable, and has virtually no overfitting risk. If interpretability were the priority, it would be the choice. The Decision Tree sacrifices some of that simplicity in exchange for capturing non-linear patterns in the data, which the higher R^2 confirms is worth it here.